

Is Dentistry Safe From AI?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

2.7

AI EXPOSURE · GENERAL DENTISTRY

where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in dental school or training, or out of it and into the job hunt. So this guide skips the "should you study dentistry" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where dentistry sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. On AI exposure, you've chosen well: chairside dentistry is among the best-protected work we score. Almost none of your clinical exposure is fixed against you. The live questions from here are financial and directional — how the debt is managed, whether the path points at ownership, and whether you land close to the chair or drift toward the tier that automates. The second half of this guide is those moves.

And one thing specific to dentistry. This is the profession where our whole framework has already been tested by history — and it passed. A tier of dentistry was automated more than a decade ago, before anyone was discussing AI, and the profession is fine. The section on the experiment dentistry already ran explains what happened and what it predicts — and why the tier you steer toward matters more than the qualification you hold.

The short answer

On AI exposure, dentistry is among the best-protected work we score.

General dentistry scores **2.7** out of 10. Oral surgery scores **2.3**. Dental hygiene scores **3.0**.

Physical work in a patient's mouth, comprehensive licensure, and a relationship built over years — that combination is close to the strongest in this index.

The reasons to think hard about dentistry are entirely financial: **average dental school debt of \$293,900**, and an income picture that depends heavily on whether someone eventually owns a practice. That financial fork — not AI — is where your decisions from here actually land.

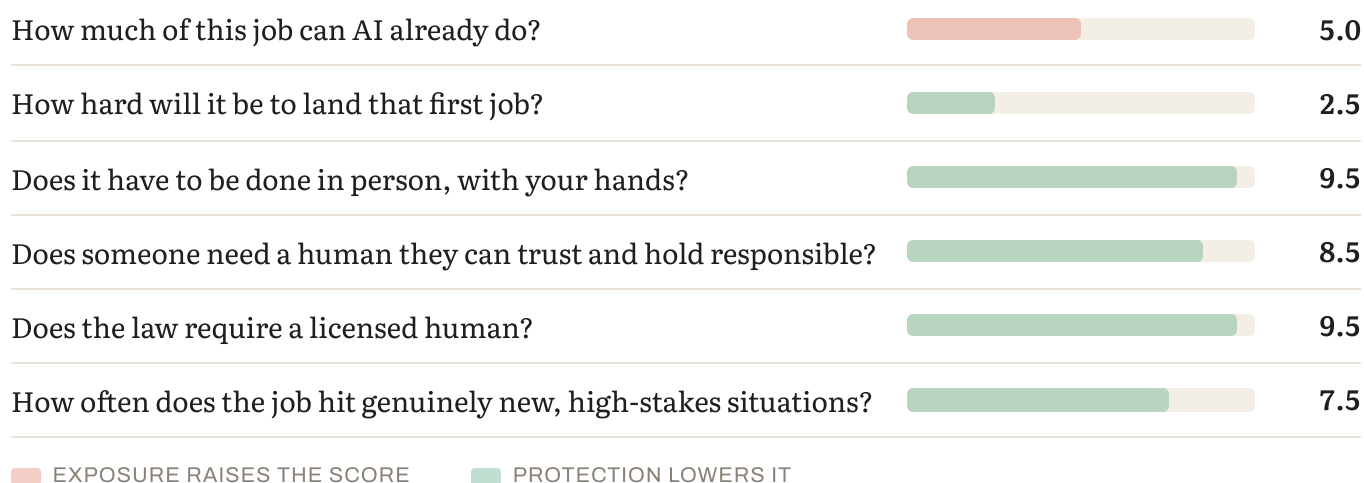
The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Oral & maxillofacial surgery	2.0	2.2	2.3	+0.3	Very low
General dentistry	2.3	2.5	2.7	+0.4	Very low
Dental hygiene	2.6	2.8	3.0	+0.4	Low
Orthodontics	3.0	3.4	3.8	+0.8	Low
Dental laboratory technology	4.8	5.2	5.5	+0.7	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, surgery scores 2.1, physical therapy 2.5, bedside nursing 2.8, and entry-level software development 8.1.

Read the table this way, and note the last row particularly — because dental laboratory work is the most instructive number in this guide, and the section on the experiment dentistry already ran explains why. Your job, from here, is to stay near the top of that table — close to the chair — not to drift down it.

Why dentistry scores so low — the six factors



Here's how general dentistry answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

Imaging support, scheduling, records, and treatment planning assistance. Not the dentistry.

Radiograph analysis and caries detection. Treatment plan drafting. Clinical notes and records. Appointment scheduling and recall. Insurance claims and coding. Restorative design and milling — which the experiment section covers, because that one already happened.

What resists: working inside a mouth with hand instruments, managing a frightened patient, judgment about what a specific tooth needs, and the relationship that brings someone back for twenty years. Those are the things to be building toward.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Strong protection. Licensure requires supervised clinical training and school places are capped, so entry is competitive but employment afterwards is secure.

The thing to steer here isn't whether you get hired — it's which tier you land in. Entry is genuinely protected, but the pattern that runs through this whole guide is that **the further the work moves from the patient's mouth, the higher the exposure climbs.** The way through is to aim at the chairside clinical seat that scores 2.7, with real procedure breadth and a path toward ownership — not to drift toward the administrative and lab-adjacent work that scores 5.5 and above just because it's the rung that happens to be open. That's a set of concrete choices, and the second half of this guide is about making them.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Almost entirely — and with unusual precision. Dentistry is fine motor work in a confined, wet, moving space, on a patient who may be anxious or in pain. This is among the harder physical problems in medicine, not among the easier ones.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and unusually so. Dentistry involves fear more than most healthcare, and trust is what gets people through the door. A licensed dentist is professionally answerable for the outcome.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Comprehensively. Practice, prescribing and procedures are all legally reserved.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Regularly. Every mouth is different, and the judgment about restoration, extraction or referral is made with incomplete information about what's under the surface.

Dentistry already ran this experiment

This is the most useful section in the guide, and it applies well beyond dentistry.

Dental laboratory technology was automated more than a decade before the AI conversation started.

For most of the twentieth century, crowns, bridges and dentures were made by hand by skilled technicians — a genuine craft, learned over years, requiring real artistry. Then CAD/CAM arrived: digital scanning, computer design, milling machines. Much of that work moved from a bench to a screen and a machine, and the number of people needed to do it fell substantially.

Notice what happened next, because it is exactly what our framework predicts.

The profession was fine. Dentists still practice. Patients still get crowns. Demand didn't fall.

But a whole tier of skilled, well-regarded craft work contracted permanently, and the people in it did not simply move up — many moved out. Dental lab technicians now report median pay around \$58,307, and the work that remains is increasingly digital design rather than manual fabrication.

And that is why lab technology still scores 5.5 today while chairside dentistry scores 2.7. The exposure isn't in front of it. Much of it has already happened.

The general lesson, and it is the single most transferable idea in this index: automation does not usually destroy a profession. It removes a tier of it — and which tier you are in matters more than which profession you choose.

Three practical consequences worth carrying to any career:

The tier that goes first is the one with **structured inputs and a physical output that a machine can produce** — a crown from a scan, a document from a template, an image from a brief.

The tier that survives is the one **with the patient, the client, or the accountability** in front of it.

And the displacement is usually invisible from outside the profession. Nobody outside dentistry noticed the lab tier contract. The same is now happening inside law, accounting and design, and it looks the same from the outside: the profession appears healthy, and one layer of it has quietly stopped being a career.

What the trend shows

General dentistry: 2.3 → 2.5 → 2.7. Up 0.4 in three years, and all of it administrative.

Orthodontics: 3.0 → 3.4 → 3.8. Up 0.8 — the fastest-moving track in the profession, driven by treatment planning, aligner design and remote monitoring being genuinely automatable.

Dental lab: 4.8 → 5.2 → 5.5. Continuing a displacement that started years ago.

The gap widened from 2.8 points to 3.2. The pattern holds exactly: the further the work moves from the patient's mouth, the higher the exposure climbs.

The financial picture — and this is the real decision

Dentistry is one of the most expensive qualifications available, and the numbers deserve careful reading — the more so once you're already carrying, or about to carry, the debt.

The debt:

- Average dental school debt is **\$293,900**, per the ADA's Survey of Dental School Seniors.
- The spread is enormous: public schools typically **\$200,000–\$260,000**, private around **\$321,000**, and the most expensive programs reach **\$625,000–\$700,000**.
- One institution's program cost alone is reported at **\$492,820**.

The income, and the gap inside it:

- BLS reports a mean annual wage of **\$166,810** for general dentists.
- ADA Health Policy Institute puts median net income for practice owners closer to **\$239,000**.
- ADA HPI analysis of Department of Education data puts average net income for general practitioners **five years after graduation at \$171,062**.
- New dentists typically earn **\$150,000–\$200,000** in their first year.
- Specialists out-earn generalists by roughly 30–90%.

That \$166,810-versus-\$239,000 gap is the whole story. Dentistry's income model assumes eventual practice ownership, and it's worth understanding that now rather than discovering it a decade into repayment.

And one statistic that says something the others don't. ADA data indicates that around **35% of dental school costs are financed without education loans**, with a further 17% covered by family. Dentistry draws disproportionately from families able to fund it, which means the "average debt" figure understates the burden for students who borrow the whole amount.

The practical read, from where you stand: the school decision may be behind you, but the ownership decision is not. The spread between the cheapest and most expensive accredited program exceeds \$400,000, which is why the debt you carry has to be planned against the income model — and that model runs through ownership, not the employed wage.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Very strong	Most fragile long-term	The protection physical AI targets — though a wet, confined, moving mouth is among the harder robotics problems.
Regulatory / licensing	Strong	Most durable	Practice and procedures are legally reserved.
Human trust / accountability	Strong	Durable	Dentistry involves fear; trust is what brings people back.
Judgment under novelty	Good	Eroding slowly	Diagnostic imaging support is advancing; chairside judgment is not.

The honest read. Dentistry has all four protections operating together, which is rare — and the historical evidence in the experiment section shows the framework working prospectively. The automation arrived where the model says it should: at the structured, machine-producible tier — which is why the direction you steer, toward the chair rather than away from it, matters more than where you happen to start.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Reading the radiograph unaided	First-pass caries detection arrives; the dentist adjudicates
Writing clinical notes	Reviewing an ambient draft
Fabricating restorations by hand	Already gone — CAD/CAM did this a decade ago
Drafting the treatment plan	Reviewing a generated plan and judging whether it fits
Insurance coding and claims	Substantially automated
Working inside the mouth	Unchanged
Managing a frightened patient	Unchanged

Human strengths that will matter most

1. **Fine motor work in a confined space** — the core of the craft, and among the hardest physical problems in medicine. 2. **Managing fear** — dentistry involves more patient anxiety than most healthcare, and handling it well is a genuine and undersupplied skill. 3. **Judgment on what a specific tooth needs** — restore, extract, refer, or wait. 4. **Long-term relationships** — the practice model depends on people coming back for decades. 5. **Verification** — knowing when an automated caries detection or treatment plan is wrong.

Notably absent: the hand fabrication of restorations — the lab-bench craft that CAD/CAM already took. That was once a respected marker of skill, and it is the clearest thing on the "already gone" list. If any part of the work you're drawn to sits on that machine-producible side, the work is to shift toward the chairside side.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The specific tools — imaging AI, chairside CAD/CAM, digital treatment-planning software — will turn over many times across a career. The dentists who compound are the ones who stay genuinely curious about the work and re-tool fastest, rather than defending the one way of working they trained on. In a profession where the digital workflow keeps moving into the operator, that flexibility is itself a form of protection — and increasingly it's what the strongest practices are built around.

Other things to consider about this job

What's genuinely good about it

Satisfaction is high and unusually consistent. Survey data puts career satisfaction around 74%, with 85% saying they would choose their specialty again — figures at the higher end of healthcare.

And the working pattern is better than most of medicine. Dentists report averaging around 36 hours a week, with no routine night or weekend work in most general practice — a genuine difference from medicine and nursing.

What dentists report as rewarding:

- **Immediate, visible results** — you fix something and see it fixed
- **Combining manual craft with clinical judgment**, which few careers offer
- **Autonomy**, with practice ownership a normal destination rather than an exotic one
- **Long relationships** with patients and families across decades
- **Genuine control over working pattern**, particularly for owners

What's genuinely hard about it

The debt, covered in the financial section, and it is the dominant issue.

The physical toll is real and under-discussed. Dentistry is performed in sustained awkward postures, and musculoskeletal problems — neck, back, shoulders, wrists — are a recognized occupational hazard that can shorten careers.

Patient anxiety is a constant. A meaningful share of patients arrive frightened, and managing that repeatedly is emotionally demanding in a way outsiders rarely appreciate.

The business is a second job. Ownership is where the income is, and running a practice means staff, overheads, insurance negotiation and marketing — none of which dental school teaches well.

And insurance economics are a persistent frustration, with reimbursement pressure a leading complaint in the profession.

Who this work suits — and who it doesn't

Dentistry tends to suit people who:

- **Have genuine manual dexterity** and enjoy precise physical work
- **Are comfortable with people who are frightened** — this is most of the interpersonal job
- **Want immediate, visible results** rather than long-horizon outcomes
- **Are interested in running a business**, or at least willing to be
- **Can work in confined, repetitive physical conditions** for decades

It's harder for people who:

- Want **variety of setting** — it's largely the same room, the same chair
- Are uncomfortable **causing discomfort**, which is unavoidable
- Have no interest in the **commercial side**, given where the income sits
- Need **intellectual novelty** daily — much of general practice is repetitive by design

If some of that second list sounds like you, it isn't a reason to abandon the path — it's a reason to steer toward the parts of the profession that fit, and toward ownership, which is where the autonomy and the income both sit.

What else is moving this market

Consolidation is reshaping the profession. Dental support organisations have grown substantially, and the proportion of dentists in traditional solo private practice has fallen. That changes the ownership assumption underpinning the income model, and it's the most important structural shift in the profession right now.

Demand is stable and demographically supported, with an ageing population retaining more natural teeth than previous generations.

And the specialty premium is significant — specialists out-earn generalists by 30–90%, at the cost of additional years and additional debt.

The AI-native advantage — what it actually looks like

The situation: dental AI is arriving as imaging support, treatment planning assistance and administrative relief. The clinical work is untouched. Being already in it, close to the work, is a real advantage here — you can build this fluency inside a live clinic rotation or first job, which is exactly where it counts.

WHAT AN AI-NATIVE DENTIST LOOKS LIKE

- 1. Verify.** Knowing when an automated caries detection is wrong in either direction. A false positive means unnecessary treatment on a healthy tooth, which is a clinical and ethical problem, not just a technical one.
- 2. Use imaging support properly.** These systems are genuinely good and genuinely confident when wrong.
- 3. Run the digital workflow.** Intraoral scanning, CAD/CAM chairside milling, digital treatment planning. The lab tier moved into the practice — a dentist who can operate that workflow captures value that used to leave the building.
- 4. Handle patient AI use.** Patients arrive having consulted symptom checkers and comparison sites, often about cost. Working with that rather than dismissing it is a new communication skill.
- 5. Take the business seriously.** Everything in the financial section points at ownership, and practice management is the least-taught and most consequential skill in the profession.

All five are available to you now — inside a current rotation, an associate seat, or a first job. Turning them into concrete moves is the rest of this guide.

Where a dental qualification can lead later

Clinical practice across general and specialist areas, practice ownership and multi-site groups, academia and teaching, public and community dental health, industry and device companies, and dental research.

The pattern in this index holds: the further from the chair, the higher the exposure. Laboratory and administrative roles score 5.5 and above against chairside practice at 2.7. That gradient is the compass for every choice below.

Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: on AI exposure, chairside dentistry is already among the safest work we score. The live spread you're steering inside runs from chairside practice (2.7) to the lab and administrative tier (5.5), and where you land is decided mostly by two things still in front of you: how close you stay to the chair, and whether the path points at ownership.

If you're still in the degree or training

Three levers, in order of leverage.

1. The ownership path is the financial whole game. The clinical work is protected on the score; the money is not automatic. The income model assumes eventual ownership — \$166,810 employed against roughly \$239,000 for owners — and dental school teaches practice management poorly. Every practice-management and business unit you can take seriously now is the highest-leverage thing left in the degree, because it's the least-taught and most consequential skill in the profession, and it's far cheaper to acquire in training than to learn on the job with a \$293,900 loan behind you.

2. The digital workflow, while it's still free. The lab tier (5.5) moved into the practice via CAD/CAM. A dentist who runs intraoral scanning, chairside milling and digital treatment planning captures value that used to leave the building — and pulls the machine-producible tier inside the protected one rather than ceding it. Get maximum exposure to scanning, milling and digital planning while your program is paying for it.

3. Clinical volume and verification, earlier than feels comfortable. The protection is the hands-in-the-mouth work (2.7). The more procedures you actually perform before graduating, the more of that protected core you own — and it varies enormously between programs, so chase every case you can. Alongside it, build the verification habit: knowing when an automated caries detection is wrong — a false positive means treating a healthy tooth is a clinical skill, not a technical one, and it's the one the imaging tools make scarcer and more valuable.

If you've graduated and you're job-hunting

Here the setting on the offer matters more than it looks, because the gradient is fixed: the further from the patient's mouth, the higher the exposure.

- **The chairside clinical seat is the 2.7.** Aim at a first role with real procedure breadth, not one weighted toward the administrative or lab-adjacent tier that scores 5.5 and above. Proximity to the chair is proximity to the protection.
- **Understand associate versus ownership economics before signing.** Ownership is where the income is; get any employment contract reviewed, and know what path to ownership — if any — the role actually offers.
- **Weigh the specialty question with eyes open.** Orthodontics is the fastest-moving track (3.8), driven by treatment planning, aligner design and remote monitoring being automatable — a 30–90% premium bought with more years and more debt.

- **Read the practice's model, not just the pay.** A seat built around routine, machine-producible or purely administrative throughput points the wrong way; one that builds clinical breadth, digital-workflow fluency and a route toward ownership points the right way.

The AI-native move, from where you stand

You can't change the tier the lab experiment already settled, but you can be the dentist who captures the digital workflow rather than watching value leave the building. The five capabilities above — verify, use imaging support properly, run the digital workflow, handle patient AI use, take the business seriously — are all available right now, inside a current rotation or first job. The most powerful for someone already in it is the third: be the dentist who runs intraoral scanning, chairside milling and digital treatment planning end to end. The lab tier moved into the practice; whoever operates that workflow captures the value that used to leave it — and you can start building that fluency this term, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in dentistry is inherited from the work itself, but the financial outcome and the tier you land in are built here, in these habits and choices. The first list is for anyone still in the degree or training; the second is for the job hunt itself.

During the degree / training — build what can't be automated

- **Treat the ownership path as the financial core.** The income model assumes eventual ownership — \$166,810 employed against roughly \$239,000 for owners — and the \$293,900 average debt has to be planned against it. Take every practice-management unit seriously; dental school teaches it poorly, so pursue it deliberately.
- **Maximise clinical volume.** The protected core (2.7) is the hands-in-the-mouth work. The more procedures you actually perform before graduating, the more of that core you own — and it varies enormously between programs, so chase it.
- **Learn the digital workflow while it's free.** Intraoral scanning, chairside CAD/CAM milling, digital treatment planning. The lab tier (5.5) moved into the practice; the dentist who runs that workflow captures value that used to leave the building.
- **Build verification into how you read imaging.** Automated caries detection is confident when wrong; knowing when to overrule it — a false positive means treating a healthy tooth — is a clinical skill, not a technical one.

- **Get exposure to a range of practice settings**, including DSO-affiliated ones, so you understand associate versus ownership economics before you're negotiating them.
- **Feed curiosity and adaptability on purpose.** The tools — imaging AI, CAD/CAM, treatment-planning software — will turn over repeatedly; the dentist who stays curious and re-tools fastest compounds.

During the job hunt — aim at the chair, don't just apply

- **Target the protected lane on purpose.** A chairside clinical seat with real procedure breadth (2.7), not a role weighted toward the administrative or lab-adjacent tier that scores 5.5 and above.
- **Read the setting, not just the offer.** The gradient is fixed: the further from the patient's mouth, the higher the exposure. A seat built around routine, machine-producible or purely administrative throughput points the wrong way.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. A practice or group still building its model around the machine-producible or administrative tier may simply be behind the curve — and that's exactly the rung that contracted in the lab experiment. Availability can be a lagging indicator, not a green light.
- **Understand associate versus ownership economics before signing.** Ownership is where the income is; get any employment contract reviewed, and know what path to ownership the role actually offers.
- **Ask the employer where AI is heading in the role.** Whether they're building the practice around the digital workflow — scanning, chairside milling, digital planning — or waiting to be forced to, tells you if the seat grows or is quietly being hollowed out.
- **Weigh trajectory over the first-year number.** A seat that builds clinical breadth, digital-workflow fluency and a path to ownership beats a higher first-year figure in a lane that's drifting away from the chair.

Go deeper — further reading

- **ADA Health Policy Institute — dental practice research** — American Dental Association. The profession's own data on income, practice ownership, consolidation and workforce. The primary source for most of the financial section, and free.

- **Dentists — Occupational Outlook Handbook** — US Bureau of Labor Statistics. Median pay, projected growth and entry requirements. Read the **Dental Hygienists** page alongside it — that comparison is the most useful twenty minutes available on this decision.
- **Average dental school debt** — program-by-program cost data, including the finding that around 35% of dental school costs are financed without loans. Worth reading before signing for any further study.

The counterargument — read this one:

The strongest case against our 2.7 isn't that AI will replace dentists. It's that dentistry already showed us what happens, and we may be under-reading it.

On this view, the lab tier's disappearance is not reassurance — it is the warning. A skilled, respected, decades-old craft inside dentistry was substantially automated within about ten years of the enabling technology maturing, and the people in it did not move up. If milling machines could do that to lab work, the argument runs, chairside robotics could eventually do something similar to routine restorative procedures, and our 2.7 reflects present capability rather than trajectory.

Where we land: we take it seriously, and it's why physical protection is the factor we rank least durable everywhere in this index. What holds our score is the specific difficulty of the environment — a wet, confined, moving mouth belonging to an anxious person is a materially harder robotics problem than a milling machine working from a scan, and the regulatory and trust protections would remain even if the manual problem were solved.

But the honest position is that dentistry demonstrates our framework works — including when the answer is uncomfortable.

Bottom line

On AI exposure, dentistry is among the safest work we measure — which is exactly why being already in it is not a problem. Chairside practice at 2.7 is protected by all four of our factors at once — fine motor work in a mouth, comprehensive licensure, deep patient trust, and clinical judgment on incomplete information.

And dentistry gives this whole index its clearest proof. The lab tier was automated more than a decade before anyone discussed AI, the profession was fine, and a tier of skilled craft work contracted permanently. Automation removes a tier, not a profession — and which tier you're

in matters more than which profession you choose. Standing still on the machine-producible tier is the one thing not to do.

The real decision is financial. Average debt of \$293,900, against a mean wage of \$166,810 that rises to around \$239,000 for practice owners. The income model assumes ownership, and steering toward it is most of what turns a protected score into a good living.

So the useful advice isn't a verdict on the degree — it's a direction. Take practice management seriously, because ownership is where the income sits and it's the least-taught skill in the profession. Learn the digital workflow, because the lab tier moved into the practice and whoever runs it captures the value. Maximise clinical volume, because the hands-in-the-mouth work is the protected core. And aim the job hunt at the chair — a seat with real procedure breadth and a path to ownership — not at the administrative or lab-adjacent tier that scores 5.5 and above.

And one honest thing about the physical protection: it's the factor we rank least durable everywhere in this index. A wet, confined, moving mouth is a hard robotics problem, and the regulatory and trust protections would hold even if that problem were solved — but the dentist who stays closest to the chair, runs the digital workflow, and owns the business is the one best placed however the technology moves.

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which track are you on — or aiming at — and does it point at *chairside clinical practice* (2.7) or toward the *administrative and lab-adjacent* tier (5.5)? 2. Looking at the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. Do you have a realistic picture of the ownership path — \$166,810 employed against roughly \$239,000 for owners — or only of the employed wage?

The move from here

4. What's the most protected, ownership-track seat you could realistically aim for from where you are now, and what would make you a credible candidate for it? 5. If you're still studying: are

you getting real clinical procedure volume and hands-on digital-workflow time, or is that being underweighted — and can you add more before it's no longer free? 6. If you're job-hunting: are you aiming at a chairside clinical seat with a path to ownership on purpose, or applying to whatever's open? Name the roles you're actually chasing.

The AI question, practically

7. Of the five capabilities — verify, use imaging support, run the digital workflow, handle patient AI use, take the business seriously — which one could you visibly demonstrate within the next few months? 8. Can you run an intraoral scan, a chairside mill and a digital treatment plan end to end? If not, where's the easiest place to start building that?

Testing it against reality

9. **The most valuable single action:** find a dentist five years qualified and ask directly about their loans, whether they'd choose the same school, and how close they are to ownership. 10. If the first offer that comes is weighted toward the administrative or lab-adjacent tier, what's your plan — take it as a deliberate stepping-stone with an exit in mind, or hold out for a chairside seat with a path to ownership? Decide before the offer, not after.

This is analysis and informed judgment about a fast-moving situation. For dentistry specifically, we've flagged that the profession's most serious risk is financial rather than technological, and that the automation of dental laboratory work more than a decade ago is the clearest historical test of the framework used throughout this index.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Dentistry

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

ORAL & MAXILLOFACIAL SURGERY — 2.3 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.2	4.6	4.8	1.68
Entry-path erosion	15%	1.2	1.5	1.8	0.27
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	2.3 → 2.3

GENERAL DENTISTRY — 2.7 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.7	5.1	5.6	1.96
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.3	9.3	9.3	0.1
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	2.69 → 2.7

DENTAL HYGIENE — 3.0 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.7	5.2	5.7	1.99
Entry-path erosion	15%	1.8	2.0	2.2	0.33
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	9.0	9.0	9.0	0.15
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	3.0 → 3.0

ORTHODONTICS — 3.8 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.6	6.7	7.7	2.69
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	8.5	8.5	8.5	0.22
Human trust / accountability	15%	9.0	9.0	9.0	0.15
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	3.79 → 3.8

DENTAL LABORATORY TECHNOLOGY — 5.5 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.5	4.4	5.1	1.78
Entry-path erosion	15%	4.5	5.0	5.5	0.82
Physical / embodied	15%	5.0	5.0	5.0	0.75
Human trust / accountability	15%	4.0	4.0	4.0	0.9
Regulatory / licensing	10%	3.0	3.0	3.0	0.7
Judgment under novelty	10%	4.5	4.5	4.5	0.55
				Total	5.51 → 5.5

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Oral & maxillofacial surgery

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	4.2	1.2	0.35	2.0
2025	4.6	1.5	0.35	2.2
Fall 2026	4.8	1.8	0.35	2.3

Movement decomposition: automatability contributed **+0.21** and entry-path erosion **+0.09**, for a total of **+0.3** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.